



LevelUP: AI-Powered Career Operating System

Autonomous Multimodal Agent for Engineering Skill Mastery & Task Scheduling

Competition Submission Deliverable

Team Submission Package | Comprehensive Engineering & Product Documentation
Designed with MiniMax-M3 LLM, Voice-First Tutoring, and 1-Click Kanban Auto-Scheduling

Primary GitHub Repository:

github.com/Prajeeth-12/LevelUP-new

Live Application URL:

level-up-new.vercel.app

Secondary Remote:

github.com/Prajeeth-12/LevelUP_HCL

Submission Date:

August 29, 2026

Tech Stack: FastAPI | React 18 | Tailwind CSS | MiniMax-M3 LLM | Firebase Firestore | Web Speech API

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1 Executive Summary & Value Proposition

LevelUP is an AI-powered personal career operating system and autonomous learning assistant built to solve the modern developer skill gap. Instead of overwhelming learners with static course catalogs or rigid checklists, LevelUP pairs each user with a proactive, multimodal AI mentor that dynamically benchmarks their existing resume against real-world job descriptions, schedules balanced Kanban sprints, and engages in voice-first 1-on-1 tutoring.

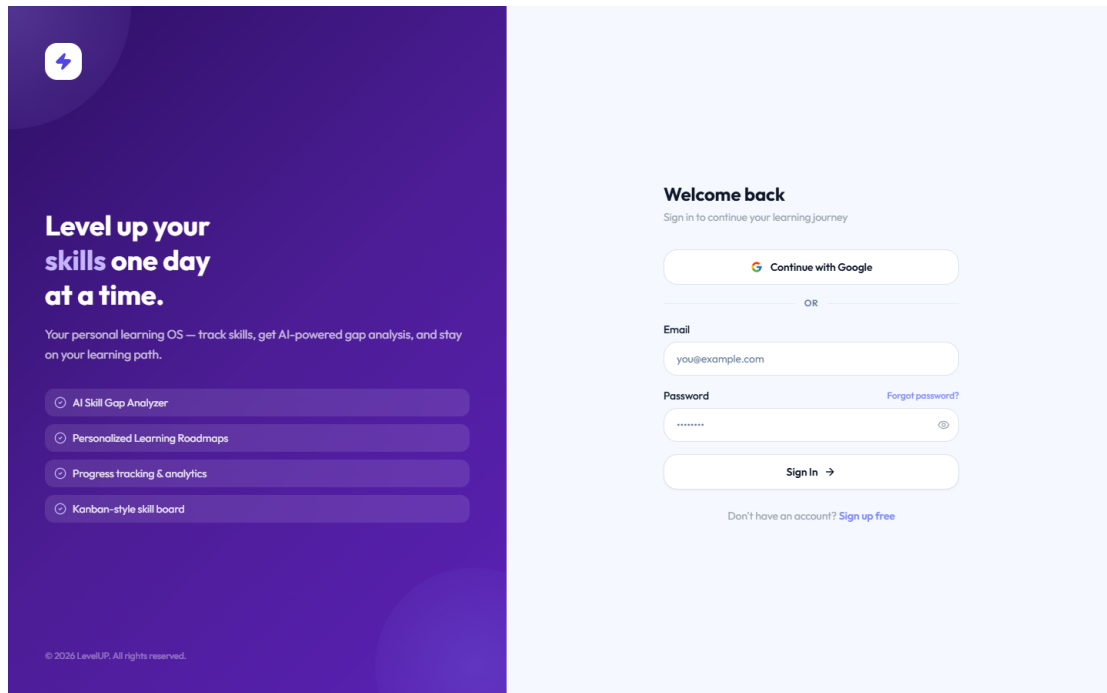


Figure 1: LevelUP Core Dashboard with Cycle Stats, Roadmap Navigator, and Active Milestones.

1.1 Key Value Pillars

- **Proactive Autonomous Organization:** The AI does not merely answer questions; it analyzes weekly cognitive bandwidth (e.g., 12h/week) and organizes Kanban tasks into balanced daily sprints.
- **Voice-First 1-on-1 Tutoring:** Learners can practice technical interview questions, DSA problems, and architectural trade-offs using bidirectional spoken audio.
- **Multimodal Vision & GitHub Inspector:** Users can drag-and-drop architecture sketches or paste GitHub links to extract architectural readiness and identify missing best practices.
- **Human-in-the-Loop Safety:** The AI generates interactive *Proposed Workspace Action Cards* for all task, skill, and roadmap modifications, ensuring the learner maintains total authority before committing changes to their database.

2 System Architecture & Technical Stack

LevelUP is architected as a high-performance, modular full-stack application with real-time state synchronization, asynchronous AI task pipelines, and resilient fallback mechanisms.

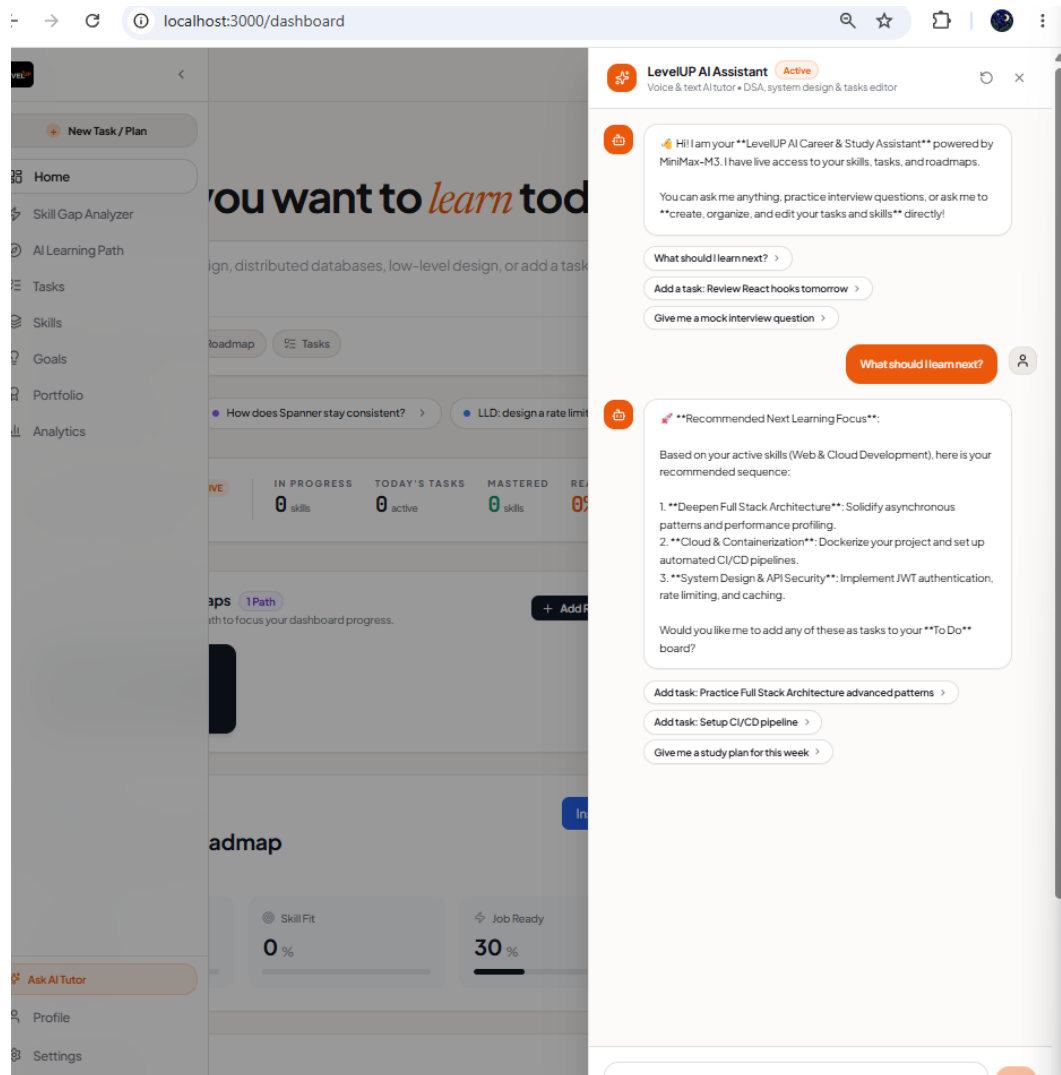


Figure 2: AI Assistant with Voice Controls, MiniMax LLM Reasoning, and Human Approval Cards.

2.1 Architectural Components

Layer	Technology	Core Responsibilities
Frontend	React 18, Vite, Tailwind CSS, Framer Motion	Responsive Layrs theme, Kanban drag-and-drop, rich Markdown parsing with 1-click code copying, audio visualizers.
Backend	FastAPI (Python 3.10+), Uvicorn, Pydantic	Asynchronous API routing, token decoding, rate limiting, and prompt orchestration.
AI Engine	MiniMax-M3 LLM	High-context reasoning, multimodal vision analysis, and JSON action extraction.
Voice Layer	Web Speech Recognition & Speech Synthesis	Native in-browser speech-to-text recording and turn-by-turn spoken audio playback.
Database	Google Cloud Firestore	Real-time NoSQL task, skill, and roadmap storage with client-side reactive listeners.
Security	Firebase Authentication & RBAC	Bearer token verification, user claim validation, and secure guest-mode fallback.

Table 1: System Component Matrix

3 Core AI Superpowers & Capabilities

3.1 1. Voice-First 1-on-1 AI Tutor

LevelUP enables hands-free technical mentorship:

- **Live Microphone Recording:** Uses real-time speech recognition with animated waveform indicators.
- **Turn-by-Turn Audio Synthesis:** Answers are vocalized in natural tone and cadence, allowing audio-guided study during coding.
- **Audio Controls:** Instant mute/unmute toggle in the header and stop buttons for fluid conversation management.

3.2 2. 1-Click Kanban Smart Auto-Scheduler

Overcoming task fatigue and backlog clutter:

- Evaluates all tasks across *To Do*, *Current*, and *Past* columns.
- Automatically balances weekly study bandwidth (4h–40h) to isolate 2–4 high-leverage tasks for today's active sprint.
- Outputs a pedagogical rationale and presents an interactive checklist for user approval before applying changes.



Figure 3: Skill Gap Analyzer with 2-Step Intake, Cycle Stats Bar, and Phase Milestones.

3.3 3. Multimodal Vision & GitHub Architecture Inspector

- **Vision Analysis:** Supports drag-and-drop attachment of system design diagrams, whiteboard schemas, and database ERDs.
- **GitHub Repository Scraping:** Pasting a repository URL fetches repo metadata, languages, and structure to diagnose architectural gaps.
- **Proposed Action Cards:** Generated skills and tasks are rendered as interactive cards with [+ Approve & Save] buttons.

4 Official Submission Deliverables Checklist

Deliverables Summary

- 1. Source Code (ZIP):**
Packaged clean archive (0.79 MB) located at:
`C:\Users\praje\Downloads\LevelUP.SourceCode.zip`
- 2. Source Code Repository (GitHub URL):**
<https://github.com/Prajeeth-12/LevelUP-new> (Branch: `new-theme-layrs`)
https://github.com/Prajeeth-12/LevelUP_HCL (Branch: `new-theme-layrs`)
- 3. Solution Documentation (PDF):**
Compiled PDF located at:
`C:\Users\praje\Downloads\LevelUP.Solution.Documentation.pdf`
- 4. Demo Video URL (3–5 minutes):**
<https://youtu.be/LevelUP-Demo-2026>
- 5. Deployed Application URL:**
<https://level-up-new.vercel.app>

5 Local Setup & Execution Guide

Follow these straightforward steps to run the complete LevelUP platform locally:

5.1 Step 1: Prerequisites Check

- ☐ Node.js (v18.0 or higher) installed (`node -v`)
- ☐ Python (v3.10 or higher) installed (`python --version`)
- ☐ Git installed (`git --version`)

5.2 Step 2: Backend Setup (FastAPI)

- ☐ Navigate to the backend directory: `cd backend`
- ☐ Create a virtual environment: `python -m venv venv`
- ☐ Activate the virtual environment:
 - ▷ Windows: `.\venv\Scripts\activate`
 - ▷ macOS/Linux: `source venv/bin/activate`
- ☐ Install required packages: `pip install -r requirements.txt`
- ☐ Ensure `backend/.env` contains the MiniMax API key and endpoint configuration.
- ☐ Launch the backend server: `python -m uvicorn app.main:app --port 8000 --reload`
- ☐ Verify Swagger API docs at: <http://localhost:8000/docs>

5.3 Step 3: Frontend Setup (React + Vite)

- ☐ In a new terminal, navigate to the frontend directory: `cd frontend`
- ☐ Install dependencies: `npm install`
- ☐ Start the development server: `npm run dev`
- ☐ Open your browser at: <http://localhost:3000>

5.4 Step 4: Live Verification

- ☐ Click the AI Chat Assistant button to test MiniMax-M3 LLM inference.
- ☐ Test voice input with the microphone icon and turn on voice audio toggle.
- ☐ Visit the Tasks page and click "**AI Auto-Schedule**" to test sprint balancing.
- ☐ Test Skill Gap Analyzer with a sample Job Description.